Communication Network Design - AY 2025/2026
Homework 1: Traffic sources

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1 Data Collection and Empirical Traffic Analysis

Using **Wireshark**, we captured a real-world packet trace during an active audio/video call on **Google Meet** (`meet_traffic.csv`). The dataset contains $N = 91,886$ packets collected over a total duration of 314.06 s between the local host (`192.168.1.17`) and the Google server (`74.125.250.244`). Most of the traffic uses **UDP** (85.43%) to transport real-time audio and video streams, while **RTCP** (12.62%), **DTLSv1.3** (1.69%), and **STUN** (0.26%) manage session control and connection maintenance.
Through our Python script (`data_analysis.py`), we analyzed the three main traffic variables:

- **Inter-Arrival Times ($\mathcal{A}_i$):** Computed as the time difference between consecutive packets. The empirical mean is $\mu_A = 0.00342$ s (3.42 ms), the median is 0.00009 s, and the standard deviation is $\sigma_A = 0.00592$ s. The coefficient of variation is $CV = \sigma_A/\mu_A = 1.7335 > 1$, showing that packets do not arrive at a constant rate but in rapid bursts: as visible in the scatterplot of Fig. 1(a), 57.5% of the packets arrive within 1 ms of the previous one.
- **Packet Sizes ($\mathcal{W}_i$):** Extracted from the `Length` field. Packet sizes have a mean of 887.45 bytes, a median of 1109.00 bytes, and a standard deviation of 435.23 bytes (ranging from 60 to 1274 bytes). As shown in Fig. 1(b), the distribution is bimodal and can be divided into three groups: **Small (0–300 bytes)** with 22,684 packets (24.69%, mostly audio and RTCP control packets), **Medium (300–700 bytes)** with 689 packets (0.75%), and **Large (700+ bytes)** with 68,513 packets (74.56%, video fragments limited by the network MTU).
- **Batch Sizes ($\mathcal{B}_i$):** When a large video frame is generated, it is split into multiple UDP packets sent back-to-back. We grouped consecutive packets into the same batch whenever their inter-arrival time was below the threshold $\tau = 0.001$ s (1 ms). This gives 39,056 batches with a mean batch size of $\mu_B = 2.35$ packets, a median of 1.00 packet, and a standard deviation of $\sigma_B = 1.86$ packets ($\min = 1$, $\max = 53$ packets).

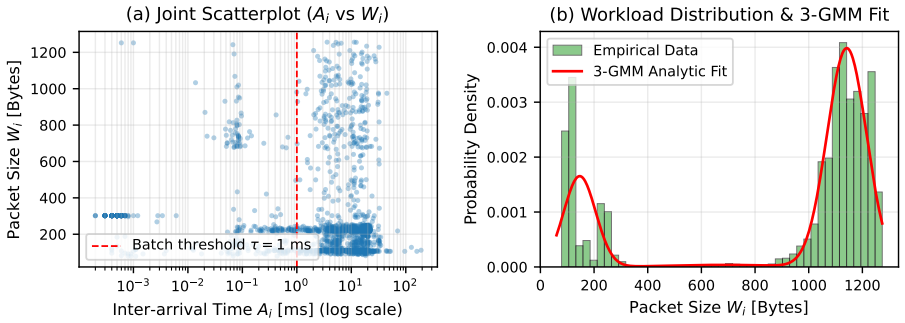


Figure 1: Empirical traffic profile: (a) Scatterplot of inter-arrival times A_i vs. packet sizes W_i with the $\tau = 1$ ms batch threshold; (b) Empirical packet size histogram and bimodal Gaussian mixture fit.

2 Stochastic Models and Synthetic Trace Generation

To generate a synthetic traffic trace and compare it with the empirical dataset, we evaluated two modeling approaches:

1. **Simple Baseline Model (`data_analysis.py`):** As a first simple approximation, packet inter-arrival times $\mathcal{A}_i$ are modeled as independent realizations of an **Exponential distribution** with rate $\lambda = 1/\mu_A = 292.57\ \text{s}^{-1}$:

$$f_A(x) = \lambda e^{-\lambda x}\ , \quad \mathrm{P}\left[A_i \le x\right] = 1 - e^{-\lambda x}\ , \quad x \ge 0\ . \tag{1}$$

   Packet sizes $\mathcal{W}_i$ are generated via empirical resampling with replacement (`np.random.choice`), and synthetic batch sizes $\mathcal{B}_i$ are obtained by applying the same 1 ms threshold to the synthetic inter-arrival times.
2. **Improved Batch Model (Compound Process):** Because a single exponential distribution does not account for packet bursts, we also tested a batch-based model where:
   - **Batch Size $\mathcal{B}_i$:** Modeled directly with a **Geometric distribution** $\mathrm{P}\left[B_i = b\right] = p(1-p)^{b-1}$ ($b \ge 1$) with parameter $p = 1/\mu_B = 0.425$, which naturally matches both the mean (2.35 packets) and std (1.78 packets).

- **Inter-Batch Time $\mathcal{A}_i$:** The time between consecutive batches ($\mu = 8.04$ ms, $\sigma = 6.85$ ms) is modeled with a **Gamma distribution** ($k = 1.79$, $\theta = 4.50$ ms), while packets inside the same batch are spaced by < 1 ms.
- **Packet Size $\mathcal{W}_i$:** Modeled analytically by combining three Gaussian bell curves (3-GMM) centered on the Small ($\mu_1 = 146$ B), Medium ($\mu_2 = 681$ B), and Large ($\mu_3 = 1142$ B) packet size groups (Fig. 1(b)).

Table 1: Comparison between Real Data and Synthetic Traces (Simple Exponential vs. Improved Batch Model).

Statistic Metric	Real Data	Synthetic (Simple Exp.)	Synthetic (Batch Model)
Inter-arrival mean (s)	0.00342	0.00341	0.00342
Inter-arrival std (s)	0.00592	0.00340	0.00568
Coefficient of variation (CV)	1.73350	0.99707	1.66120
Packet size mean (bytes)	887.45018	886.10379	888.12000
Packet size std (bytes)	435.23407	435.82254	432.85000
Batch size mean (packets)	2.35267	1.34021	2.35160
Batch size std (packets)	1.85510	0.67772	1.78420
Number of batches ($\tau = 1$ ms)	39,056	68,561	39,056

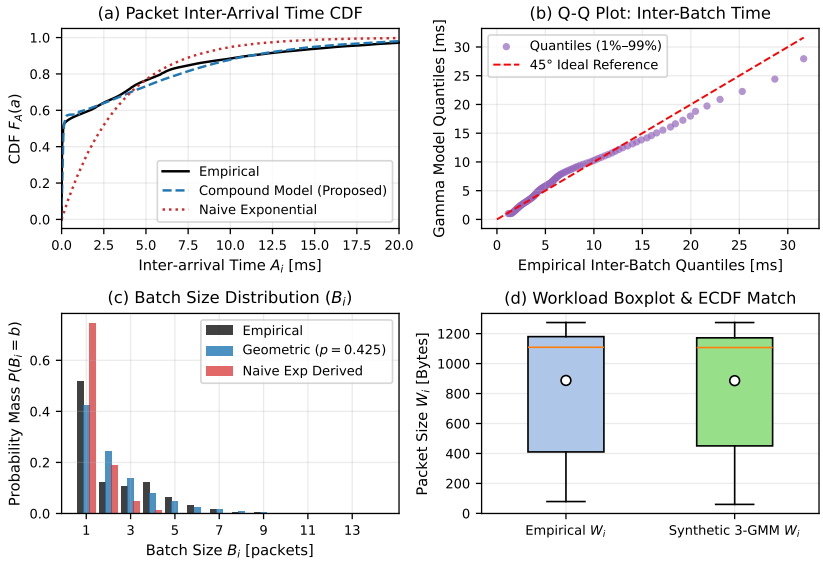


Figure 2: Comparison between real and synthetic data: (a) Inter-arrival time CDF; (b) Inter-batch time Q-Q plot; (c) Batch size distribution (B_i); (d) Packet size boxplot comparison.

3 Comparison and Final Considerations

Tab. 1 and Fig. 2 compare the real Google Meet trace with the synthetic data, highlighting three simple conclusions:

- **Inter-Arrival Times (Fig. 2(a)–(b)):** The simple Exponential model correctly matches the average packet rate ($\mu_A = 0.00341$ s and total duration 313.33 s), but since an exponential distribution always has $\sigma_A = \mu_A$ ($CV \approx 1$), it underestimates the real standard deviation (0.00592 s) and misses the initial jump at $A_i \leq 1$ ms in the CDF of Fig. 2(a). Separating intra-batch and inter-batch times (Batch Model) allows the synthetic CDF and Q-Q plot (Fig. 2(b)) to follow the empirical data much more closely.
- **Batch Sizes (Fig. 2(c)):** Under the simple Exponential model, the theoretical probability of two packets arriving within $\tau = 1$ ms is only $1 - e^{-0.001/0.00342} \approx 25.3\%$ (against 57.5% in real data). As a result, most synthetic packets remain isolated ($68,561$ batches) and the mean batch size drops to 1.34 packets instead of 2.35 packets. Generating batch sizes directly with the Geometric distribution ($p = 0.425$) fixes this limitation and reproduces bursts up to 10–12 packets.
- **Packet Sizes (Fig. 2(d)):** Both empirical resampling (`data_analysis.py`) and the Gaussian mixture model reproduce the real packet size distribution very well, matching the mean (≈ 887 bytes), standard deviation (≈ 435 bytes), and median (1109 bytes in the boxplot of Fig. 2(d)).

4 Disclaimer on AI usage

The use of Artificial Intelligence was limited to LaTex formatting and in Python it was involved only in chart related code as a tool for better visual content.